

House Price Prediction – Week 1

Internship Project

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Project Title: House Price Prediction

Project Overview

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property features such as area, number of bedrooms, bathrooms, stories, parking availability, air conditioning, and furnishing status. The Housing Prices Dataset from Kaggle was used for this analysis.

Data Exploration and Cleaning

The dataset contained 545 rows and 13 columns. The target variable was **price**, while the remaining columns were used as input features. The dataset was checked for missing values and duplicate records. No missing values were found, and duplicate records were removed. Categorical variables such as mainroad, guestroom, basement, airconditioning, and furnishingstatus were converted into numerical format using one-hot encoding.

Model Development

Two machine learning regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regression

The dataset was split into 80% training data and 20% testing data. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Model Performance

Linear Regression

- MAE: 970,043
- RMSE: 1,324,507

- R^2 Score: 0.653

Random Forest Regression

- MAE: 1,021,546
- RMSE: 1,400,566
- R^2 Score: 0.612

The results indicate that Linear Regression performed slightly better than Random Forest Regression on this dataset.

Important Features Affecting House Prices

Feature importance analysis showed that the following factors had the greatest influence on house prices:

1. Area
2. Number of Bathrooms
3. Air Conditioning
4. Parking Availability
5. Number of Stories
6. Number of Bedrooms

Among all features, the area of the house had the highest impact on the predicted price.

Interesting Observations

Several interesting insights were discovered during the analysis:

- Larger houses generally had significantly higher prices.
- Houses with more bathrooms tended to be more expensive.
- Air conditioning positively influenced house prices.
- Properties with parking facilities and multiple stories were usually priced higher.
- Although Random Forest is often expected to outperform Linear Regression, Linear Regression achieved a better R^2 score on this dataset.
- House prices were not evenly distributed and showed a right-skewed distribution, indicating that a small number of houses were much more expensive than the majority.

Conclusion

The project successfully demonstrated how machine learning can be used to predict house prices using property characteristics. Linear Regression achieved the best performance with an R^2 score of 0.653, meaning the model explained approximately 65% of the variation in house

prices. The analysis revealed that area, bathrooms, air conditioning, parking, and stories are key factors influencing house prices.

Recommendation

Real estate businesses should focus on properties with larger living areas, modern amenities, adequate parking, and air conditioning facilities, as these features contribute significantly to higher property values and better market demand.